



Birla Institute of Technology & Science, Pilani
Pilani Campus
ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2025-26

Course Handout Part II

Date: 1st August 2026

Course Number : CE F417
Course Title : Applications of Artificial Intelligence in Civil Engineering
Instructor-in-charge : RAJIV GUPTA

Course description: Introduction to Artificial Intelligence and soft computing; Neural networks: Introduction, models, and its application in civil engineering, fuzzy logic and its application in decision making, Clustering; Genetic algorithms; Expert systems; Introduction to machine learning (Support Vector Machine), case studies.

Scope and Objective: Artificial Intelligence (AI) may be considered a field of computer science that attempts to build technology to inculcate human cognition in computer systems, but applications are not limited to. A primary goal of AI is to build intelligent entities. This course is structured to give an overview of the area, as well as some important soft computing techniques which are essential to solve the complex problems in terms of the fundamentals. The attributes of human intelligence such as reasoning, planning, learning etc. will be attempted to be understood to incorporate in problemsolving.

Course learning outcomes:

- By the end of the course, the students should have a. Out of various techniques, emphasis will be on Genetic algorithm, Artificial Neural Network, and Fuzzy Systems
- General knowledge of the field of AI
 - They should be able to recognize when and how to use AI techniques to solve civil engineering problems.
 - The students should also be able to evaluate new techniques they encounter.
 - Students will take up a group project/ term paper to study in detail the theoretical details of an intelligent system and implement that during the semester.

Text books and references:

TB1: Artificial Intelligence A Modern Approach; Stuart J. Russell, Peter Norvig, Third Edition Pearson Education, Inc., India
TB2: Neural Networks, Fuzzy logic, and Genetic Algorithms Synthesis and Applications; S. Rajasekaran, G.A. Vijayalakshmi Pai, PHI Learning Pvt. Ltd, Delhi.

Method of conduct of the course:

The traditional method of teaching by chalk and board has been highly regarded for many educational levels. The chalk-and-board teaching system has been appreciated for creating an opportunity for direct and personal communication between teachers and students to engage with each other face to face. Approaches such as blended learning, collaborative strategies, and experiential learning promote engagement, critical thinking, and lifelong learning.

Course Plan: weekly schedule for 16 weeks with [3] lectures per week

Week no.	Module	Lect. no.	Reference	Learning Outcome
1.	Overview of AI	1	T1	Application of AI
		2	T1	Introduction to AI
2.		3-4	Class notes	Projects and Implementation
2.		5	T1	Forms of Learning

3.	Artificial Neural Network	6	T2	Supervised Learning
		7-8	T2	Regression and Classification with Linear Models
4.		9 -11	Class notes	Application and problem solving
5.	Fuzzy System	12-14	T2	Fuzzy Set theory
6 - 7		15-16	T2	Fuzzy Systems
		17-19	Class notes	Applications and problem solving
8-10	Genetic Algorithm	20-21	T2	Fundamentals of GA
		22-24	T2	GA Modeling
		25-28	Class notes	Applications and problem solving
11-12	KBS	29	T1	Logical Agents First order Logic
		30-32	T1	Knowledge Representation
13-15	Hybrid System	33-35	T2	GA based Back Propagation Network
		36-38	T2	Fuzzy Back Propagation Networks
		39-41	Self	Application, Implementation of Civil Engineering projects
16		42	Self	Presentation of Projects

Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
Problem solving	20%	30 Class + 20	Once in a week	Buffer 20%
Project	30%	Continuous	Title, Group, Methodology, Ms, Final	All components are must
Webinar	15%	90 min *4	Once in a month	3/4
Comprehensive	35%	180 min	13/12 FN	OB

Criterion for NC:

1. As per academic regulation
2. Final Project Marks will be awarded only if the student completes project as per the course requirement.
3. Missing any component completely will lead to NC

2. Chamber consultation hours: TH- 9th Hour

3. Makeup Policy: Prior permission is must.